



UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

FIRST SEMESTER EXAMINATION, APRIL 2023

COURSE NO: MC/EL/GM/GL/ES/CE 361

COURSE NAME: PROBABILITY AND STATISTICS

CLASS: MC/EL/GM/GL/ES/CE III

TIME: 3 HOURS

Name: _____

Index Number: _____

INSTRUCTION: *ANSWER ALL QUESTIONS BY CIRCLING THE RIGHT ANSWER ON THE QUESTION PAPER. EACH QUESTION ANSWERED CORRECTLY ATTRACTS ONE MARK.*

1. Which of the following mentioned standard Probability density functions is applicable to discrete Random Variables?
 - a. Normal Distribution
 - b. Binomial Distribution
 - c. Exponential Distribution
 - d. Rayleigh Distribution
2. What is the probability of an impossible event?
 - a. 0
 - b. 1
 - c. *Not defined*
 - d. *Insufficient data*
3. A variable that can assume distinct values is called
 - a. Continuous random variable
 - b. Discrete random variable
 - c. Irregular random variable
 - d. Random variable
4. Which one of the following variables is a continuous random variable?
 - a. The time it takes a randomly selected student to complete an exam
 - b. The number of shareholders of a randomly selected company

- c. The number of correct guesses in a multiple choice test
 - d. The number of students in a class
5. A stick of unit length is broken randomly into three pieces. (Specifically, two points a ; b are chosen at random on the stick and it is cut at these two points.) What is the probability that the three stick pieces can be formed into a triangle?
- a. $\frac{1}{4}$
 - b. $\frac{1}{2}$
 - c. 1
 - d. $\frac{1}{8}$
6. If A and B are two mutually exclusive events with $P(A) = 0$ and $P(B) = 0$ then it implies they are also independent
- a. True
 - b. False
7. Let A and B be two events such that $P(A) = \frac{1}{5}$ While $P(A \text{ or } B) = \frac{1}{2}$. For what values of P are A and B independent?
- a. $\frac{1}{4}$
 - b. $\frac{1}{2}$
 - c. 0, *since mutually exclusive*
 - d. $\frac{5}{18}$
8. A sample of 35 observations has the mean 80 and standard deviation 4. A second sample of 65 observations from the same population has mean 70 and standard deviation 3. The standard deviation of the combined sample is
- a. 5.85
 - b. 37.2
 - c. 5.58
 - d. None of these
9. Two sisters maintain that they can communicate telepathically. To test this assertion, you place the sisters in separate rooms and show sister A, a series of cards. Each card is equally

likely to depict either a circle or a star or a square. For each card presented to sister A, sister B writes down 'circle', or 'star' or 'square', depending on what she believes sister A to be looking at. If ten cards are shown, what is the probability that sister B correctly matches at least one?

- a. 0.9910
- b. 0.9827
- c. 0.5111
- d. 0.9100

10. The variance of the 20 observations is 5. If each observation is multiplied by 2, find the variance of the resulting observations

- a. 10
- b. 20
- c. 40
- d. 5

11. A process for putting chocolate chips into cookies is random and the number of choc chips in a cookie has a Poisson distribution with mean λ . Find an expression for the probability that a cookie contains less than 3 choc chips.

- a. $P(X < 3) = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2} \right)$
- b. $P(X < 3) = e^{-\lambda} \left(1 + \lambda^3 + \frac{\lambda^3}{3} \right)$
- c. $P(X > 3) = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^4}{4} \right)$
- d. $P(X \geq 3) = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2!} \right)$

12. A manufacturer produces piston rings for an automobile engine. It is known that ring diameter is normally distributed with $\sigma = 0.001$ millimeter. A random sample of 15 rings has a mean diameter of $\bar{x} = 74.036$ millimeters. Construct a 99 % two-sided confidence interval on the mean piston ring diameter.

- a. $74.0355 \leq \mu \leq 74.0367$
- b. $72.0911 \leq \mu \leq 70.0111$
- c. $60.0011 \leq \mu \leq 61.2231$

d. $74.1123 \leq \mu \leq 74.9810$

13. Suppose n balls are distributed at random into r boxes. Find the probability that there are exactly k balls in the first $r_1 (< r)$ boxes.

- a. $\binom{n}{k} \left(\frac{r_1}{r}\right)^k \left(\frac{r_1}{r}\right)^{n-k}$
- b. $\binom{n}{k} \left(\frac{r_1}{r}\right)^k \left(1 - \frac{r_1}{r}\right)^{n-k}$
- c. $\binom{n}{k} \left(1 - \frac{r_1}{r}\right)^k \left(\frac{r_1}{r}\right)^{n-k}$
- d. $\left(1 - \frac{n}{k}\right)^k \left(\frac{r_1}{r}\right)^k \left(\frac{r_1}{r}\right)^{n-k}$

14. A regression analysis yields the line

$$\hat{y} = 18 + 0.25x$$

One of the subjects, Mary, has $x = 40$ and $y = 32$. Calculate Mary's residual.

- a. -6
 - b. -2
 - c. -4
 - d. -8
15. During the course of an experiment with a particular brand of tractor engines, it is found that 80% of the tractor engines are damaged in the first experiment. The tractor engines that survived developed significant amount of wear and tear, so that the percentage of engines that survived in any later experiment is half that in the preceding experiment. Find the probability that an engine will survive 4 experiments.
- a. 0.08
 - b. 0.05
 - c. 0.45
 - d. 0.11
16. The shape of a normal curve is?
- a. Bell shaped
 - b. Flat

- c. Circular
- d. Spiked

17. A statement whose validity is tested on the basis of a sample is called?

- a. Null Hypothesis
- b. Statistical Hypothesis
- c. Simple Hypothesis
- d. Composite Hypothesis

18. Let X be a random variable on with probability mass function (PMF) $f(x) = cx^2$ defined on $x = 0, 1, 2$.

- a. 3
- b. 1
- c. c
- d. $1/3$

19. What does correlation coefficient compute?

- a. The strength of associations between data variables
- b. The strength of associations between two variables
- c. The strength of associations between independent variables
- d. None of the above

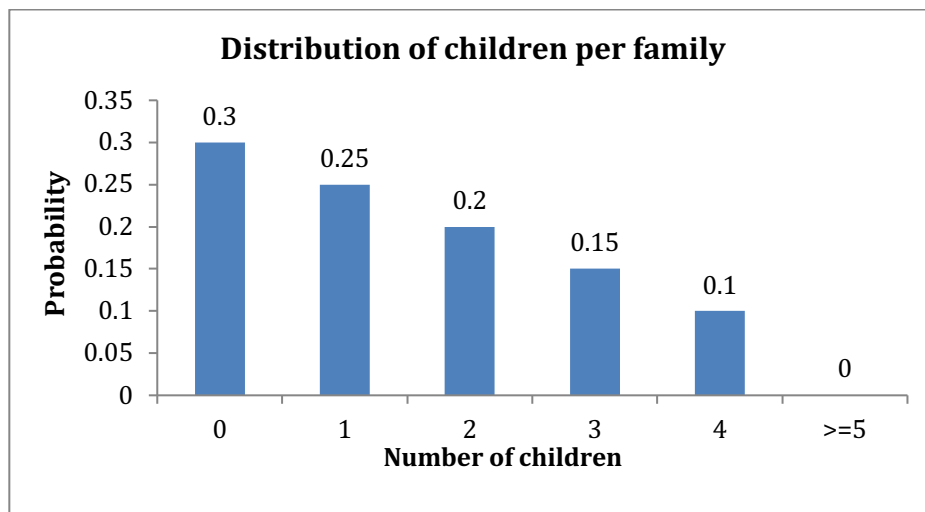
20. A research on “Near Surface Characteristics of Concrete: Intrinsic Permeability”, presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. Summary quantities are $n = 8$, $\sum x = 43$, $\sum y = 572$, $\sum x^2 = 157.42$, $\sum y^2 = 23,530$ and $\sum xy = 1697.80$. Assume that the two variables are related according to the simple linear regression model. Which of the simple regression model fits the summary quantities?

- a. $\hat{y} = -28.90 + 18.68x$
- b. $\hat{y} = -28.90 - 18.68x$
- c. $\hat{y} = 28.90 - 18.68x$
- d. $\hat{y} = -28.90x - 18.68$

SECTION B
ATTEMPT TWO (2) QUESTIONS ONLY

Question 1 (20 marks)

- a. Assume the distribution of children per family is as given in the graph below:



- i. Consider a random girl and her brother in the population of children. What is the probability that she has a brother? *Hint: Here the girl is at random (9 marks)*
 - ii. What type of random variable does this distribution follow? *(1 mark)*
- b. A bag contains fifteen (15) balls distinguishable only by their colours; ten (10) are blue and five (5) are red. I reach into the bag with both hands and pull out two balls (one with each hand) and record their colours.
- i. What is the random experiment? *(1 mark)*
 - ii. What is the sample space? *(1 mark)*
 - iii. Express the event that the ball in my left hand is red as a subset of the sample space. *(2 marks)*
- c. Given that 633.8 and 160.8 gives the mean and standard deviation of a sorted data with $n = 5$ from a population. Calculate 90% confidence interval for the mean of the population. *(3 marks)*
- d. Suppose a continuous random variable X is defined on the normal distribution; State the properties of X . *(3 marks)*

Question 2 (20 marks)

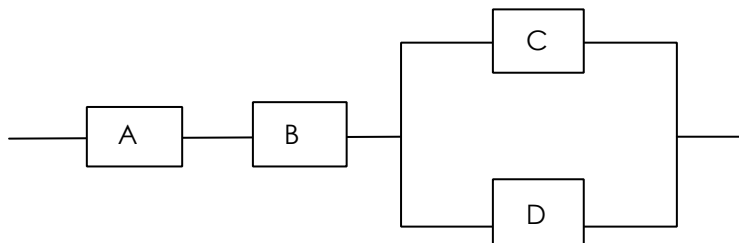
- a. An interviewer is given a list of 8 people whom he can attempt to interview. He is required to interview exactly 5 people. If each person (independently) agrees to be interviewed with probability $\frac{2}{3}$. What is the probability that his list will enable him to complete his task? **(5 marks)**
- b. The distance, in kilometres, travelled to work by the employees of a city council may be modelled by a normal distribution with mean 7.5 and standard deviation 2.5. Find the probability that the distance travelled to work by a randomly selected employee of the city council is;
- Less than 11.0 km. **(2 marks)**
 - Between 5.5 km and 10.5 km. **(4 marks)**
 - Find d such that 10% of the council's employees travel more than d km to work. **(2 marks)**
 - Find the probability that the mean distance travelled to work by a random sample of 6 cleaners of the council's employees is less than 5.0 km. **(2 marks)**
 - Without further calculation**, comment on the fact that the mean distance travelled to work by the 6 people employed to clean the council's offices is 4.4 km. **(2 marks)**
- c. Verify that the probability distribution function is a probability density function (PDF)

$$f(x) = \frac{1}{21}(x + 1)$$

where $x = 1, 2, 3$. **(3 marks)**

Question 3 (20 marks)

- a. On each of four days next week (Monday to Thursday), Dan will shoot five free throws. Assume that Dan's shots satisfy the assumptions of Bernoulli trials with probability of a failures $q = 0.34$.
- Compute the probability that on any particular day Dan obtains exactly three successes. For future reference, if Dan obtains exactly three successes on any particular day, then we say that the event "**Win**" has occurred. **(1 marks)**
 - Refer to part (i). Compute the probability that: next week **Win** will occur exactly once and that one occurrence will be on Wednesday. **(3 marks)**
- b. An electrical system consists of four components as illustrated in the figure below.



The system works if components A and B work and either of the components C or D work. It is known that the components work independently and that $P(A \text{ works}) = P(B \text{ works}) = 0.9$ and $P(C \text{ works}) = P(D \text{ works}) = 0.8$.

Find the probability that;

- the entire system works, and **(3 marks)**
- the component C does not work, given that the entire system works. **(2 marks)**

- c. A manufacturer of auto windows has developed a new plastic protective material that can be applied much thinner than the conventional material. To use this material, however, the production machinery must be adjusted. A trial adjustment was made on one of the 10 machines used in production, and a sample of 25 windshields measured. This sample had a mean thickness of 2.9 mm . Using the population standard deviation of 0.25 mm , does this adjustment provide for a smaller thickness in the material than the old adjustment (4 mm)? (Use a hypothesis test and level of significance of 0.01. Assume the distribution of thickness is approximately normal.) **(8 marks)**
- d. Give a counter example to show that pairwise independence of a set of events $A_1, A_2, A_3, \dots, A_n$ does not imply mutual independence. **(3 marks)**